

CONTACT

Isfahan, Iran

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TECHNICAL SKILLS

Programming

C · Embedded C · C++ · Python

MCUs

STM32 (F4/F7) · ESP32 / ESP32-S3 · ESP8266

Frameworks & Tools

STM32CubeMX/IDE · Keil MDK · ESP-IDF · Git · Doxygen

Protocols & Peripherals

UART · I²C · RS-485 · Modbus RTU · MQTT · DMA · Timer input capture

Debug & Test

ST-LINK · Logic analyzer · Hardware bring-up · Simulators

RTOS

FreeRTOS

AI / ML

PyTorch · OpenCV (in progress) · ESP-DL · Familiarity with machine learning and deep learning concepts

LANGUAGES

Persian	native
English	upper-intermediate (B2)
Chinese	elementary (≈HSK 2)

Mohammad Soroush Rabiei

Embedded Systems Developer

SUMMARY

Embedded systems developer with industrial experience in data-center infrastructure management (DCIM). My work and interests span IoT and connected devices, edge AI, and building smart products — mostly at the boundary of hardware and software.

EXPERIENCE

Embedded Systems Developer — Jul 2026 – Present
Kian Pardaz Nagh-e Jahan · Isfahan, Iran

- Researched and selected temperature/humidity sensors for the DCIM product.
- STM32 driver development :
  - For the DHT22 temp/hum sensor (Non-Blocking State Machine + Timer Input Capture + DMA)
    https://github.com/MohammadSoroushRabiei/STM32-DHT22-Driver
  - For the Modbus RTU master over the RS-485 communication channel
    https://github.com/MohammadSoroushRabiei/STM32-Modbus-RTU

Embedded Systems Intern — Jun 2025 – Sep 2025
Behbood Ertebat Chehelsotoon Co. (Behamad) · Isfahan, Iran

- Developed STM32 firmware for a smart-city (municipal IoT) device with LoRaWAN and a SIM800 GSM module.
- Implemented MQTT on STM32 to connect a smart thermostat to the platform.

PROJECTS

Smart Home — Self-Hosted Edge-AI Platform · B.Sc. final project
github.com/MohammadSoroushRabiei/Smart-Home

- A cloud-free smart home on ESP32-S3: on-device face recognition (ESP-DL), a self-learning on-chip behavior agent, an LVGL touch dashboard and a fully local MQTT/Home Assistant stack.

IoT Smart Room · IoT course project
github.com/MohammadSoroushRabiei/IoT_SmartRoom_ESP8266

- Remote room control and monitoring (lights, fan, temperature) on ESP8266 with Blynk.

CERTIFICATES

PCB Design with Altium Designer (Essentials) — Oct 2026
Certificate of Completion · FaraDars



Certificate of Participation — IoT 2025 — Oct 2025
The 9th International Conference on Internet of Things and Its Application · University of Isfahan

EDUCATION

B.Sc. in Computer Engineering — 2020 – 2026
Intelligent Systems · Isfahan University of Technology

Diploma, Mathematics & Physics — 2018
NODET (SAMPAD) High School, Isfahan